Professional Installation Manual

Enterprise Server Deployment and Operating System Installation

Course: ITEP 414 – System Administration and Maintenance

Program: Bachelor of Science in Information Technology (BSIT)

Project: Week 3 Portfolio Project

Name: Allyn Jade L. Delgado

Section: BSIT-4B

Server: Ubuntu Server

Hostname: server01

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1. Introduction

This manual presents a practical procedure for setting up an Ubuntu Server virtual machine in an enterprise-style environment. The server can later support file sharing, remote administration, databases, web hosting, and other internal applications. The steps are organized so another administrator can repeat the setup and obtain a similar result.

2. Objectives

The main goals of this deployment are to:

Install Ubuntu Server inside a virtual machine.

Apply the required server and network settings.

Set the hostname and create an administrative user.

Configure storage using the guided disk option.

Install and enable OpenSSH Server.

Check the hostname and assigned IP address.

Confirm Internet connectivity.

Update the operating system.

Confirm that the SSH service is active.

Record the installation and verification process.

3. Software Requirements

Prepare the following before starting:

Oracle VirtualBox or VMware Workstation

Ubuntu Server LTS ISO

A computer capable of running the selected virtual machine

Windows Server Evaluation ISO for the separate bring-home activity, if required

4. Virtual Machine Specifications

Component	Configuration
VM Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB

Disk Format	VDI or VMDK
Network	NAT or Bridged if instructed
Optical Drive	Ubuntu Server ISO
Hostname	server01

These specifications are based on the requirements for the Week 3 deployment activity.

5. Ubuntu Server Installation Guide

Step 1 – Create the Virtual Machine

Open VirtualBox or VMware Workstation and create a VM named Ubuntu-Server-Week03. Assign 4 GB RAM, 2 virtual processors, a 40 GB virtual disk, NAT networking, and mount the Ubuntu Server ISO. Add a screenshot of the completed VM settings.

Step 2 – Select Language

Boot the VM from the Ubuntu Server ISO and choose English. Capture the language-selection screen.

Step 3 – Configure the Keyboard

Choose the keyboard layout that matches the computer. English (US) may be used as an example. Capture the selected layout.

Step 4 – Configure the Network

Allow Ubuntu Server to obtain its network settings through DHCP unless another configuration is required. Record the assigned IP address and include a screenshot.

Step 5 – Configure the Hostname

Set the hostname to server01. This name will identify the machine on the network. Capture the hostname screen.

Step 6 – Create the Administrative User

Create a non-root administrator account. Full Name: Allyn Jade L. Delgado. Username: yourlastname. Password: use a strong password. Use this account for routine administration instead of logging in directly as root.

Step 7 – Configure Disk Partitioning

Select Guided – Use Entire Disk. Record the partition scheme, file system, and disk size. For this setup, the disk size is 40 GB.

Step 8 – Install OpenSSH Server

When the installer asks about SSH, select Install OpenSSH Server. SSH allows secure remote administration.

Step 9 – Complete Installation

Finish the installation, restart the VM, remove the installation ISO when prompted, and allow the system to boot from the installed disk. Capture the final login screen.

6. Initial Server Configuration

After logging into Ubuntu Server, perform the following checks and setup tasks.

6.1 Verify the Hostname

Run:

hostname

The expected hostname is server01. Add a screenshot of the terminal output.

6.2 Verify the IP Address

Run:

ip addr

Find the active network interface and record the assigned IP address. Add a screenshot.

6.3 Test Internet Connectivity

Run:

ping -c 4 google.com

Successful replies indicate that the server can reach the Internet. Save a screenshot.

6.4 Update the Server

Run:

sudo apt update
sudo apt upgrade -y

Wait for the update and upgrade processes to finish, then capture the result.

6.5 Verify SSH

Run:

systemctl status ssh

The service should show active (running). Capture the terminal output.

7. Server Verification Procedures

Verification Item	Expected Result	Status
Login	User can log in successfully	☐
Hostname	server01	☐
IP Address	Valid assigned IP	☐
Internet	Successful ping	☐
System Update	Updates completed	☐
SSH	Active and running	☐

Screenshots should be attached as evidence for each major verification step.

8. Troubleshooting Guide

Problem 1 – No Internet Connection

Check the VM network adapter, confirm NAT or the required network mode, run ip addr, verify an IP address was assigned, and test again with ping -c 4 google.com.

Problem 2 – Hostname Is Incorrect

Check the current hostname with hostname and confirm that it is set to server01.

Problem 3 – SSH Is Not Running

Check with systemctl status ssh. If needed, install it with sudo apt install openssh-server and check the service again.

Problem 4 – System Updates Fail

Check connectivity with ping -c 4 google.com, then retry sudo apt update followed by sudo apt upgrade -y.

9. Best Practices

Use a non-root administrative account for regular system management.

Protect administrative accounts with strong passwords.

Keep Ubuntu updated using appropriate package-management commands.

Enable SSH when secure remote administration is needed.

Document configuration changes for easier troubleshooting.

Keep screenshots and installation records as evidence.

Check network connectivity and important services after installation.

Keep server documentation organized.

Use consistent hostnames and configuration standards.

Review security settings regularly before production use.